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DEPARTMENT: EMERGING TECHNOLOGIES

Immersive Computing: What to Expect in a Decade?

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Computing technology is advancing at an unprecedented speed, reshaping both our daily lives and the global landscape. In this article, we delve into the profound impact of immersive technology on society, focusing on how it can represent and interact with users. Starting from the current virtual reality (VR) technology, we envision that VR will integrate with artificial intelligence (AI) and robotics technologies, and eventually achieve a personal clone of individual users that can perform most of the tasks on behalf of users. By roughly dividing the progression of immersive technologies into three phases, we discuss their prospective applications while emphasizing the potential research challenges to be addressed.

On 2 February 2024, the debut of Vision Pro from Apple made it to the news headlines. Vision Pro headsets can enable immersive experiences by allowing users to interact with the digital environment in a more intuitive way. It garnered widespread attention as people believe it has the potential to transform many industries, including education, training, health care, entertainment, and so forth. As a matter of fact, many have suggested that a new era of spatial computing is coming and have drawn parallels to the first generation iPhone, released in 2007, suggesting a potential life-transforming technique.

Although its debut represents a significant leap forward in the spatial computing industry, Vision Pro does not come without problems. The wave of the device returns for reasons including high price, limited applications, heavy weight, discomfort experience, and isolation also made to the news headlines around the expiration of a two-week free trial period, suggesting that there is much to improve. These controversies have added to the long-lasting debate regarding the realism of the metaverse,[1] a term that was coined roughly 30 years ago but drew attention when Facebook announced its name change to Meta in 2021.

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Although the metaverse embraces the grand vision of digital worlds that can be enabled via virtual reality, augmented reality, and mixed reality [(VR)/(AR)/(MR), also termed *extended reality* (*XR*)] techniques, undoubtedly, headsets consist of an indispensable part of the entire ecosystem.

What led to more debates during this course includes the techniques of digital twins, artificial intelligence (AI), and robotics. On one hand, digital twins are often implemented via VR/AR/MR techniques, based on a real-world model, in the hope of simulating and predicting the event's progress in the real world for manufacturing, safety, traffic control, and others. On the other hand, today, AI techniques allow much of the digital content generated without using any real-world models, but simply based on the user's text inputs. Moreover, AI-powered generative agents,[2] being entirely virtual or deployed on real robots, can book flight tickets, clean the house, and have demonstrated the capability of cooking.

Although there is no consensus regarding when and how the metaverse can be realized, blockchain is often believed to be a vital tool to bridge the digital and physical worlds,[3] via which transactions can be conducted, and interactions between the two worlds can be integrated underneath, while preserving privacy and security. Nevertheless, the fast development of AI-driven robotics can also be integrated with XR to bridge the gap between the virtual and physical worlds, realizing a more promising future.

To unravel these seemingly unrelated and sometimes conflicting and fast-paced changes while the development and interweaving of different technologies continue, as shown in Figure 1, in this article, we discuss what immersive computing is, where it will lead us, and the challenges during this course. By roughly dividing the technology advances into three levels that correspond to three development phases, we discuss the potential applications and challenges in each phase. These levels are mainly dependent on how tightly VR/AR/MR technologies are integrated with AI and robotics to represent and interact with users. Note that these phases may overlap with each other. We offer our perspectives in the hope of motivating more in-depth discussion and research, contributing to immersive computing.

WHAT IS IMMERSIVE COMPUTING?

We define immersive computing as *technologies that create highly interactive and engaging experiences for users by seamlessly integrating digital and physical world information into their physical and digital environments*. By this definition, we hope to make it clear that a user may immerse into a virtual environment (as traditionally thought) or immerse into another physical environment with the assistance of immersive technology.

As such, we expect that immersive computing will undergo at least three development phases corresponding to three levels, from personal representation (PR) to personal assistant (PA), and finally to personal clone (PC). We discuss their characteristics, exemplar applications, and challenges in the following sections.

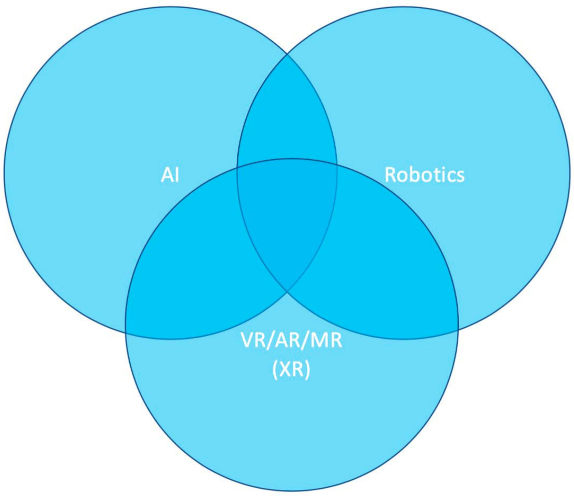

FIGURE 1. The interweaving of VR/AR/MR with AI and robotics.

LEVEL 1: PERSONAL REPRESENTATION (PR)

For the first (and current) level, immersive computing is at a stage to perform personal (or self) representation. This is an advancement from the traditional approach, such as online chatting, where users do not see each other.

In this stage, the representation of a user is typically an avatar—entirely virtual. The avatar is often not based on the user's physical (e.g., body) information, but an imaginary face, such as some cartoons, or a face with a torso, with no or a limited number of emojis.

In this stage, the avatar's actions are often driven by the user directly. That is, a user moves the controller/mouse to make the avatar walk or take other actions, and the interactions are with the user behind the avatar. Moreover, robots are rarely used to represent the users in this stage. At a scale, this forms the basis of digital twins,[5] where the actions of users in the real world are mirrored by the corresponding counterparts in the physical world. Digital twins can go beyond individual users. Based on real-world data, the digital world can use simulations to perform predictions and assist decision making, such as for traffic control (e.g., a smart city), safety (e.g., autonomous driving), manufacturing and engineering (e.g., building and bridge weakness), and medical applications (e.g., a new drug or device testing). For these applications, we note that although they share some research challenges as we discuss later, they often face other challenges in areas other than immersive technology, and we do not specifically focus on such additional challenges in this article.

Exemplar Applications

Figure 2 depicts a situation where a PR can help a working mom. This mom has to travel. In the hotel room, she uses an app on her smartphone to tell a bedtime story to her daughter until she falls asleep. Some applications of level 1 (L1) immersive technology may include the following.

- *Social VR*: Traditional VR experience is mainly for individuals. With L1 immersiveness, applications such as social VR are becoming popular. In the same virtual space, users can interact and communicate with each other "face to face," facilitating better communication and

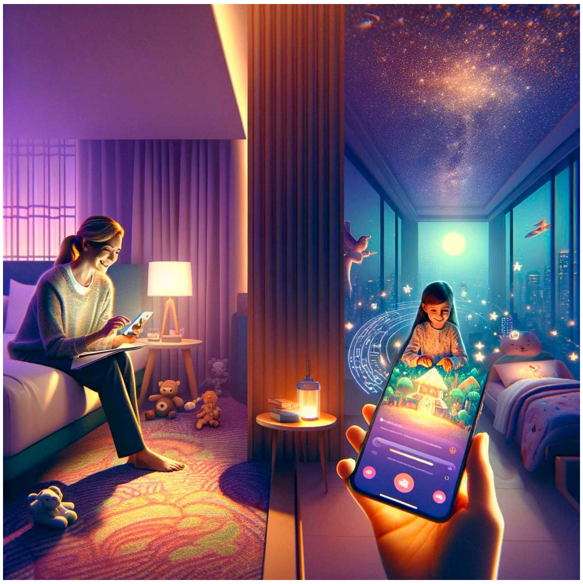

FIGURE 2. PR. A traveling mom reads a bedtime story to her daughter via a VR app on her smartphone. (Some components of this figure were generated using DALL-E.)

collaboration experiences among them. Popular social VR applications today include Meta Horizon Worlds, VRChat, and so on.

- *VR games*: Games are another popular application of immersive technology. With the capability of immersing into the virtual environment, users gain a more realistic experience when interacting with characters and objects in the game. Roblox is one of the most popular platforms for VR games.
- *Online business, such as shopping, meetings, and collaborations*: For example, with VR support, users could gain a better clothes-shopping experience using a VR fitting room. Similarly, compared to traditional Zoom meetings and collaboration, VR-based platforms, such as JoinXR, will likely provide a more engaging experience for users.

Research Challenges

For better self-representation, many challenges exist.

- *Avatar development*: Designing realistic and engaging avatars is essential as they represent users. Although it is easily created, a virtual avatar, if it doesn't reflect the physical characteristics, may lack attraction to general users. An avatar that cannot be used to express the emotions of users is also insufficient.

The development challenges lie in at least the following aspects:

  a) *Visual effects or visual realism*: Computer graphics are needed to create a realistic appearance of skin, facial expressions, and so forth.
  b) *Animation or movement*: Properly animating avatars naturally and expressing their emotions accurately based on all the surroundings as well as psychology is challenging.
  c) *Reactions*: Avatars are created to represent users in the physical world, and they need to be capable of properly reacting to the actions of other users' avatars, such as gestures, text, and speech, in real time. The development of avatars should also take into account the uncanny valley effect[4] where the increase in the human likeness of an avatar may cause uneasy or eerie feelings in viewers.

- *Action model*: An action model guides the selected actions of the avatar, e.g., the action taken by the avatar at each time step. It may refer to moving toward a direction, picking up an item, or performing certain operations. The action model could be closely related to the avatar's development, and/or the action model could be based on reinforcement learning or other machine learning (ML)/AI technology. But at this level (L1), the avatar is usually driven by the end user directly without using a model, and the user and the avatar are closely coupled.
- *Content*: Content is the key to retaining users. Creating compelling content is subject to a number of challenges depending on the application context.
  a) For social VR, the environment is very important. This requires detailed and as realistic as possible environment settings, with rich texture, proper lights, surrounding sound, and so on.
  b) For VR-based games, storytelling is the key to capturing users' attention and driving their engagement. Creating attractive characters and compelling story arcs require in-depth study.
  c) For online business, such as meetings and collaborations, the interactive model should encourage users to actively engage in the virtual environment. Fortunately, with the fast development of large language

models (LLMs) and/or other transformer-based AI models, such as Sora, this problem is expected to be mitigated greatly.

- *Hardware*: The development of hardware, particularly head-mounted displays (HMDs) or glasses, continues to be a challenge. An improvement is needed with regard to
  a) *Price and affordability*: The high cost of acquiring HMDs/glasses and relevant accessories undoubtedly hinders the wide adoption of the devices, which is one of the major complaints about Apple Vision Pro. Reducing the price while maintaining the same performance and quality in the future hardware is a must.
  b) *Comfort*: Another major reason for returning Vision Pro is because users found it uncomfortable to wear for an extended period of time. Successfully addressing the issues related to weight, motion sickness, and isolation could attract more users.
  c) *Tracking/controller*: Hands and the head are the current major tracking targets. Improving the tracking performance, particularly the accuracy within controllers or controller-free settings, is important to improve the user-perceived immersive experience.
  d) *Battery lifetime*: User convenience necessitates mobility, which requires an untethered HMD. This subsequently demands a longer battery life that doesn't need to frequently recharge, which would negatively affect usability and user experience.
  e) *System*: Most of the current immersive applications are deployed on a small scale. With the grand vision of metaverse and so on, a system capable of supporting tens of thousands of concurrent users should not be uncommon.

However, the current practice shows severe scalability issues from multiple perspectives:[6]

  a) *Bandwidth scalability*: With multiple participants, the system needs to provide support to enable real-time interactions among users. A rudimentary implementation would utilize one-to-one communication between user pairs. However, when the scale goes up, individual participant's bandwidth requirement goes up linearly, which can quickly reach the physical and practical bandwidth constraints of users.
  b) *Computing*: The scale of the participants not only challenges the bandwidth availability of the network that connects users from geographically distributed regions but also challenges the computing capability of local devices to update the content in the environment in real time based on user interactions. For example, when multiple participants move using different actions, such updates should be reflected on the display of each participant involved instantly. Omniverse[10] from Nvidia provides a potential platform and may shed some light on this issue. However, current HMDs often are not equipped with sufficient computing capability due to considerations of weight, battery consumption, and so on.

- *Security and privacy*: In immersive environments, the users' data, such as biometric and behavior actions, may be continuously collected and processed to meet the application requirements. These, however, pose a number of security and privacy concerns, not exclusively, including
  a) *Data security*: User data, such as personal information, biometric data, and behavior traces, contain sensitive information about a particular user, and thus, protecting such data is of paramount importance to immersive applications.
  b) In addition to users' sensitive data, as mentioned earlier, the content and information shared in the application among the participants may also carry sensitive information, via which an adversary may infer private and sensitive information. Furthermore, such content may be created with copyrights. Thus, properly protecting such content is also essential.
  c) *Cyberattacks*: Immersive devices are ultimately computers equipped with various sensors. They are also subject to various cyberattacks, such as phishing scams, malware infections, and distributed denial-of-service attacks. These attacks can disrupt user experience in these applications and undermine the integrity of the immersive environment.
- *Harassment, safety, and ethics*: Different from security and privacy, users in an immersive environment may be subject to sexual harassment, cyberbullying, and so forth, which can affect the safety of the participants. Although a number of cases have been reported, our

understanding of the characteristics of such behaviors is still in an early stage, let alone the appropriate detection and even prevention/protection mechanisms.[7] Therefore, in this aspect, more research is necessary to accomplish the following:

a) Understand the characteristics of harassment behaviors, such as the corresponding gestures, cultural background, and other context information, which is essential for detection and protection.
b) Design effective mechanisms to detect harassment behaviors. Currently, the detection, if one exists, is mostly manual and relies on the participant to report it. An automated mechanism can greatly improve the success rate of detection.
c) Deploy appropriate prevention and protection with automatic monitoring and moderation. Protection and prevention are the ultimate goal, particularly for the youth. Compared to detection and punishment, this would be more effective for user protection. This, such as appropriate parental control, is desirable but demands more in-depth research.

- *Interoperability*: Immersive systems, such as the metaverse, have many different implementations. However, currently, a user in one system cannot transparently participate and immerse in another system if the user wishes to. This degrades user experience. Enabling interoperability requires efforts on standardization, protocol engineering, benchmarking, and so on. Such interoperability also requires sophisticated underlying system support by transparently and seamlessly migrating the user's settings and experience from one environment to another. Web 3.0 may have the potential to provide support for these requirements.

LEVEL 2: PERSONAL ASSISTANT (PA)

In the second stage of immersive computing, the immersive technology will weave ML/AI and robotics technology.

By integrating ML/AI, the avatar in the virtual world is likely to be generated based on the full body model of the user (i.e., full embodiment), and the action model of the avatar may not only be driven by the real user in the physical world but could also be driven by the (ML/AI-based) behavior model of the user. That is, the decoupling of the user in the physical world and the corresponding avatar in the virtual world would emerge. The avatar can make independent decisions automatically without waiting for the intervention of the user in the physical world. We consider these to be personal virtual assistants (PVAs).

By also integrating robotics technology, the functions of PVAs can be greatly extended. That is, robots can be driven by PVAs to help users complete tasks while the user in the physical world is away. For example, after communicating with the household robots when it is 5 p.m. in the afternoon, the PVA may instruct the cooking robot to prepare the dinner. In this case, we refer to the cooking robot as a *personal physical assistant* (*PPA*). Compared to PVAs, PPAs have limited mobility and thus may be constrained in terms of application scope.

PAs still require synchronization with the user in the physical world periodically. On the one hand, they should get the results (of assigned tasks) back and update the status to the user. On the other hand, the user may need to adjust the behavior of the PAs, e.g., by retraining their behavior models via continuous learning and improvement.

Exemplar Applications

Figure 3 illustrates a similar situation to that of Figure 2. The working mom has to travel. In the hotel, she uses an app on her smartphone to activate a stationary robot at home to tell a bedtime story to her daughter till she falls asleep. PAs, including both PVA and PPA, have broad applications.

- *Routine tasks*: Both PVAs and PPAs can assist users with many routine tasks, as such tasks

FIGURE 3. Example of a PPA. The traveling mom uses a home robot to tell the bedtime story to her daughter. (Some components of this figure are generated using DALL-E.)

are likely to be more deterministic. For example, a PVA can book a flight without having to involve the user step by step. A PPA can help clean the house after the user leaves home and prepare meals before the user returns.

- *Tele-x*: Telepresence will be more accepted by common users. With lifelike avatars, users gain better experience in meetings and collaborations. With PVAs and PPAs, telemedicine, telecare, and telehealth will gain great popularity with reduced health-care costs. Students can also immerse into Mars, for instance, to conduct experiments for their homework assignments.
- *Accessibility support*: In addition to the help with routine tasks and tele-x, PAs can also greatly expand user accessibility. For example, it can support accessibility by using voice control for hand-free operations, using screen reading support for the visually impaired, and so forth.

Research Challenges

The development of PAs needs to address the following several challenges:

- *Avatar development*: In this stage, the requirement for an avatar representing the user is much higher. A full body-based embodiment is needed. To do so, we need to capture the characteristics of the user's entire body and movement in real time. Several challenges stem from such requirements:
  a) *Motion capture*: Capturing the user's entire body and movement requires more sensors, cameras, and/or other sensing devices that can track the position and orientation of users.
  b) *Representing*: Capturing is just the first step. Such information needs to be used to support avatar interaction and animation in a natural fashion, particularly on some unexpected events, for example, when avatars collide with each other. A realistic and responsive action is expected.
- *Action model*: In this stage, the user is partially (to mostly) decoupled from the actions of the avatar. Different from L1, wherein the user drives the avatar in real time, in L2, the action model is driven by the user's model, which may be generated based on the historical information of user behaviors, i.e., the user's behavior model. Such a model is relatively straightforward if the avatar does not need to interact with others (other virtual objects or avatars). However, when the interactions are likely while the user in the physical world is absent, the action model needs to provide sufficient support to enable such interactions.
- *Content*: In a PVA, the virtual environment demands higher resolution, rich texture, and realistic animation. With the continuous improvement of LLM-based AI tools, such as Sora[8] for text to video and DALL-E[9] for text to image, we would expect that soon, similar tools will emerge that can be used to generate 3-D content for the virtual world. On the other hand, as a PPA interacts with the physical world, the content should be a proper reflection of the physical world.
- *Hardware*: In L2, the hardware demand, on the one hand, comes from the sensors that are required to capture the full-body motion of users in real time. On the other hand, such data should be fed into the user model and drive the avatar in the virtual environment, which demands more powerful computing capability at the local device. In addition, network latency should be extremely low to deliver a realistic response.
- *System*: In L2, with a better representation of the avatar, better content, and more complicated and agile action models, the system's scalability is likely to be a bottleneck. Not only does the bandwidth for the network connecting different participants or the computing capability on the local device demand a more scalable architecture, so too does the system. Existing systems mostly utilize a client–server architecture, which is inherently not scalable on a large number of participants. Although clouds can help by elastically adding the servers upon the increase of the demand, a more scalable architecture, such as peer-to-peer based, is more desirable to connect a large number of users across the world.
- *Security and privacy*: As PAs are more capable and in many aspects act on behalf of users, security and privacy demands only increase, becoming more challenging. Additional attack vectors are likely to emerge, such as poisoning the behavior model of the users, or camouflage and impersonation attacks via avatars or models. Thus, continuous authentication should be in place as the first defense frontier. A zero-trust model has the potential to shield many of such emergent attacks.[11]
- *Harassment, safety, and ethics*: PVAs and PPAs, if not managed properly, can endanger

users. As they can more actively interact with the physical world, the damage could be severe. A PVA could harass other users in the virtual world, and a PPA can make unexpected moves in the physical world. Thus, a safety bound (e.g., a framework) should be developed to constrain and regulate the behaviors of PVAs and PPAs while the user is absent.

- *Interoperability*: The diversity of virtual environments and robotics will inevitably lead to some compatibility and interoperability issues. For protection and/or other considerations, some may be closed systems while others may be open sourced. This is likely to create barriers to interoperability among different systems. How to improve the interoperability remains a challenge to be addressed.

LEVEL 3: PERSONAL CLONE (PC)

In this stage, the avatar representing the user is further decoupled from the user in the physical world and can self-drive and make decisions by itself. In the virtual world, this requires a hologram avatar that precisely characterizes the user, and an action model that understands the user and the physical world. In this stage, we consider them PCs.[a] In the virtual world, they will be personal virtual clones (PVCs).

Based on the user model and the physical world model, a PVC may instruct the robots to complete real-world tasks, or the PVC may execute on the robots, sometimes referred to as *fully embodied AI*, which we call *personal physical clones* (*PPCs*).

In the PC stage, the PC can be entirely decoupled from the user in the physical world.

Exemplar Applications

Figure 4 shows an example of using L3 immersive technology. Like before, the working mom has to travel. When she leaves home, her full-body AI-equipped robot acts on her behalf and tells a bedtime story to her daughter until she falls asleep. L3 immersive technology will enable an unprecedented world.

- *Physi-digi reality*: A PC can help users to accomplish most of the tasks in their daily life. Users and PCs are interchangeable in most tasks, realizing a truly "mixed" and seamlessly integrated reality, which we call *physi-digi reality*.

[a]The scope of our discussion in this paper does not include the biology clone technology.

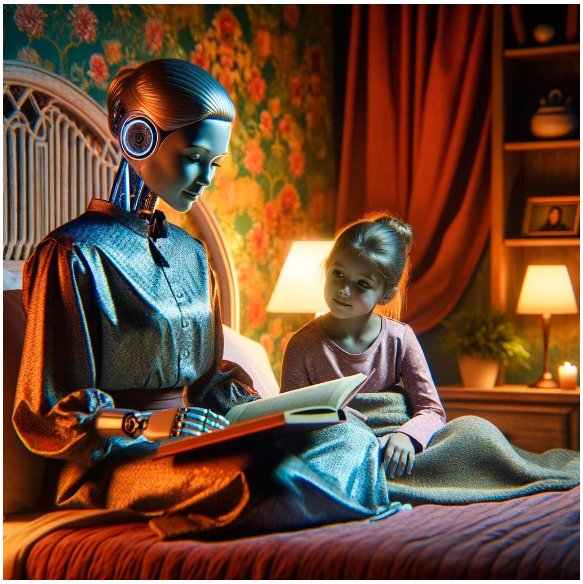

FIGURE 4. A PPC, where a home robot not only has the same appearance as the mom, but also behaves the same. (Some components of this figure were generated using DALL-E.)

- *Reincarnation reality*: With L3 immersive technology, one could be able to live in the virtual world entirely after death, which is also referred to as *digital man*, or even interact with the real world via a PPC.
- *Futureality*: Traveling to the future has been in books, movies, and TV. With PCs, particularly with the help of PVCs, L3 immersive technology can enable users to immerse into the future to some extent.

Research Challenges

Achieving L3 immersiveness is the hardest due to the following reasons:

- *Avatar development*: In L3, the avatar is critical for a PVC but may not be required for a PPC. In this stage, 3-D hologram avatars are expected. For hologram avatars, not only are realistic motion capture and natural representation in real time important but also the proper expression of body language, including facial expressions. Holographic representation is needed so that the avatar is lifelike. Photorealistic rendering must take all factors, such as light reflection, into account.
- *Action model*: In this stage, the avatar should be able to "duplicate" the behavior of the user for an extended period of time, and, ideally, for

an infinite duration. This requires a behavior model that can accurately capture the user's history and predict the user's response and action upon any event. Although ML and AI can help, proper interaction requires some reinforcement learning based on the understanding of the physical world, which is likely to take years to achieve.

- *Content*: All virtual content must support six degrees of freedom (6 DoF),[12] which is capable of supporting the users to view from any angle and perspective. Users should not be able to differentiate between the virtual content generated from the content that is a reflection of the physical world. Recent advances in the computer vision community have introduced novel 3-D representations of real-world scenes, including neural-based representations such as neural radiance fields[13] and point-based representations of 3-D Gaussian splatting.[14] These representations support the synthesis of photorealistic novel views and have the potential to revolutionize how 6-DoF contents are created and represented in the future.
- *Hardware*: The holographic display will be a key component. Such a display should be able to project the virtual content into the space using light or other optical techniques. In addition to the traditional visual display for the vision sense, other displays, such as olfactory and gustatory displays for smell and taste senses, respectively, are also needed. That is, users should have more than just the traditional visual and audio experience. These challenges could be addressed with the advances of multisensory media research.[15]
- *System*: For holographic content, bandwidth and latency requirements will be even higher. High-speed and low-latency networking is required to deliver such an immersive experience. Considering the large numbers of data transmitted, new optimizations, such as data coding and compression, could be sought.
- *Security and privacy*: A holographic avatar demands more detailed information about the user's body and behaviors. As in this stage, the PC can independently act on behalf of the user, so the security and privacy are of paramount importance. A security and privacy framework must be in place. This applies to both PVCs and PPCs. For PPCs, additional safety considerations are critical. Full-body AI-equipped robots, if misused or hacked, can lead to disasters. The challenge includes how to authenticate itself, prevent it from being attacked/hacked, and validate the future actions to be taken, and so on.
- *Harassment, safety, and ethics*: Related to security and privacy, PVCs and PPCs may result in great safety concerns. In addition to a security framework, a reliability model is also demanded to make sure all the actions taken by PCs fall into some safe boundaries. On the other hand, PCs have wide applications. How to prevent user addiction could be a potential issue in the long run.
- *Interoperability*: In this stage, the PCs much resemble the user in the physical world and should be able to immerse in and out of a lot of diverse systems that can interact with each other seamlessly.

CONCLUSION

In recent years, the rapid advancement of computing technology in VR/AR/MR, AI, and robotics has begun to revolutionize our daily lives. In this article, we focused on immersive computing that blends various technologies together and discussed the technological advances and challenges in different stages of its progression. Our primary focus is on identifying the roadblocks that hinder realization of the vision of immersive computing. We hope this article can spark increased and more profound discussions and research endeavors in this domain.

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